

Universal Vertices Force the Central Spectral Spike in Dense Kernel-Based Random Graphs

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Status: candidate full affirmative resolution, likely valid, of the simulation conjecture in arXiv:2502.09415v2. Vertices above an explicit Pareto threshold are universal and force an exact high-multiplicity eigenvalue. Their binomial count predicts about 477 forced eigenvalues in the source’s $N = 5000$ plot. This rigorously confirms the high-connectivity mechanism; it does not classify every eigenvalue in the finite-width plotting bin.

1 Source and exact conjecture

Cipriani, Hazra, Malhotra, and Salvi study a random graph on the discrete d -torus

$$V_N = \{1, \dots, N\}^d, \quad n_N := |V_N| = N^d,$$

with independent Pareto weights satisfying

$$\mathbb{P}(W > t) = t^{-(\tau-1)} \quad (t \geq 1)$$

and conditional edge probabilities

$$p_{ij} = \min \left\{ \frac{\kappa_\sigma(W_i, W_j)}{\|i - j\|^\alpha}, 1 \right\}, \quad \kappa_\sigma(u, v) = (u \vee v)(u \wedge v)^\sigma.$$

There are no self-loops. Their scaled adjacency matrix is $A_N = \mathbb{A}_{G_N} / \sqrt{c_N}$, where

$$c_N = \frac{1}{N^d} \sum_{i \neq j} \|i - j\|^{-\alpha} \sim c_0 N^{d-\alpha}.$$

The right panel of their Figure 1 has a conspicuous central spike for small α . Comparing it with a Gaussianized simulation at $\alpha = 0$, they conjecture that the spike is caused by high connectivity and by the saturation $p_{ij} = 1$ that becomes complete at $\alpha = 0$ [1].

Remark 4.3), the a.s. limiting ESD of $P_N G_N P_N$ is $\mu_{sc} \boxtimes \mu_W$. All matrices are centred and rescaled by the sample second moment. Thirdly, to elucidate the tail behaviour of the limiting ESD when $\sigma = 1$ (Theorem 2.2) we draw in Figure 3 the empirical survival function of the eigenvalues of a matrix of size 7000×7000 in $x \geq 1.5$.

Finally, we provide in Figure 4 a simulation of the eigenvalues of the Gaussian matrix $\tilde{\mathbf{A}}_{N,m,g}$ (see (4.12)) when $\alpha = 0$ and $N = 6000$. We compare this picture with the right-hand side of Figure 1, which has a small α . We conjecture that the atom appearing in the latter is due to high connectivity of the kernel-based realization (if $\alpha = 0$, for all i, j we have that p_{ij} is identically one in (2.3)), whilst in the Gaussian setup this trivialization does not arise.

The next theorem proves this mechanism directly and quantitatively.

2 Forced atom theorem

Let

$$D_N := \max_{i,j \in V_N} \|i - j\| = d \lfloor N/2 \rfloor, \quad \beta := \tau - 1,$$

and define the saturated high-weight set

$$H_N := \{i \in V_N : W_i \geq D_N^\alpha\}, \quad M_N := |H_N|.$$

Theorem 1 (Universal-vertex atom and its size). *Assume $\sigma > 0$. Then, conditional on the weights, every vertex of H_N is universal: it is joined to every other vertex with probability one. Consequently,*

$$\dim \ker(\mathbb{A}_{G_N} + I) \geq (M_N - 1)_+,$$

and the scaled matrix A_N has the exact eigenvalue

$$-c_N^{-1/2}$$

with multiplicity at least $(M_N - 1)_+$. Moreover,

$$M_N \sim \text{Bin}(N^d, D_N^{-\alpha\beta}).$$

If $\alpha\beta < d$, then, in probability,

$$M_N \sim N^d D_N^{-\alpha\beta} \sim (d/2)^{-\alpha\beta} N^{d-\alpha\beta},$$

so

$$\text{ESD}(A_N)(\{-c_N^{-1/2}\}) \geq \frac{(M_N - 1)_+}{N^d} \sim D_N^{-\alpha\beta}.$$

If $\alpha\beta = d$, then M_N converges in law to a Poisson random variable of mean $(2/d)^d$; if $\alpha\beta > d$, then $M_N \rightarrow 0$ in probability.

Proof. Fix $i \in H_N$ and $j \neq i$. Because all weights are at least one and $\sigma > 0$,

$$\kappa_\sigma(W_i, W_j) = (W_i \vee W_j)(W_i \wedge W_j)^\sigma \geq W_i.$$

Also $\|i - j\|^\alpha \leq D_N^\alpha \leq W_i$, so $p_{ij} = 1$. Thus i is universal.

Consider the coordinate subspace

$$E_N = \left\{ x \in \mathbb{R}^{V_N} : \text{supp } x \subseteq H_N, \sum_{i \in H_N} x_i = 0 \right\}.$$

It has dimension $(M_N - 1)_+$. If $x \in E_N$ and $i \in H_N$, universality gives

$$(\mathbb{A}_{G_N} x)_i = \sum_{j \in H_N \setminus \{i\}} x_j = -x_i.$$

If $i \notin H_N$, every vertex of H_N is still joined to i , and hence $(\mathbb{A}_{G_N} x)_i = \sum_{j \in H_N} x_j = 0 = -x_i$. Therefore $\mathbb{A}_{G_N} x = -x$ on E_N , proving the spectral claim.

The Pareto tail and independence give the asserted binomial law. Its mean is

$$\lambda_N = N^d D_N^{-\alpha\beta} \sim (d/2)^{-\alpha\beta} N^{d-\alpha\beta}.$$

When $\alpha\beta < d$, $\lambda_N \rightarrow \infty$ and $\text{Var}(M_N) \leq \lambda_N$, so Chebyshev's inequality yields $M_N/\lambda_N \rightarrow 1$ in probability. At equality, the standard binomial Poisson limit applies because the success probability tends to zero and the mean tends to $(2/d)^d$. Above equality, Markov's inequality and $\lambda_N \rightarrow 0$ give $M_N \rightarrow 0$ in probability. $\square$

Remark 2 (Why there is no contradiction with absolute continuity). *For every fixed $\alpha > 0$, the forced mass is of order $N^{-\alpha(\tau-1)}$ and therefore vanishes. At the same time, $c_N^{-1/2} \asymp N^{-(d-\alpha)/2}$, so the forced eigenvalue approaches zero. The theorem describes a large finite-size spike, not an atom in the limiting ESD. It is fully consistent with the source's absolutely continuous limit.*

3 The complete endpoint and Gaussian contrast

Corollary 3 ($\alpha = 0$). Put $n = N^d$. At $\alpha = 0$ the KBRG is the complete graph K_n , and

$$\text{ESD}(A_N) = \frac{1}{n} \delta_{\sqrt{n-1}} + \left(1 - \frac{1}{n}\right) \delta_{-1/\sqrt{n-1}}.$$

Thus the high-connectivity atom has multiplicity $n - 1$ at the endpoint.

Proof. Since $\kappa_\sigma(W_i, W_j) \geq 1$, every p_{ij} equals one. The spectrum of the adjacency matrix of K_n is $n - 1$ once and -1 with multiplicity $n - 1$, while $c_N = n - 1$. $\square$

Proposition 4 (No analogous forced atom after Gaussianization). *Condition on a Gaussianized variance profile in which every off-diagonal variance is strictly positive. The resulting symmetric Gaussian matrix has simple spectrum almost surely. In particular, it has no deterministic high-multiplicity eigenvalue corresponding to the universal-vertex block.*

Proof. Conditional on the profile, the independent upper-triangular coordinates have a joint density. A repeated eigenvalue is equivalent to the vanishing of the characteristic polynomial's discriminant, a polynomial in those coordinates. This polynomial is not identically zero: setting the off-diagonal entries to those of a weighted path (and all other allowed coordinates to zero) gives a real symmetric tridiagonal matrix with simple spectrum. A nonzero polynomial has a Lebesgue-null zero set. $\square$

The positivity qualification is deliberate: an artificial truncation that sets some profile entries to zero can itself create deterministic zero rows. The untruncated $\alpha = 0$ Gaussian comparison in the conjectural mechanism has a positive profile.

4 Numerical prediction for the source figure

The right-hand panel of source Figure 1 uses

$$d = 1, \quad N = 5000, \quad \alpha = 0.1, \quad \tau = 4, \quad \sigma = 1.5.$$

Thus $D_N = 2500$ and $\alpha(\tau - 1) = 0.3$, so Theorem 1 gives

$$\mathbb{E}M_N = 5000 \cdot 2500^{-0.3} = 478.176249895 \dots$$

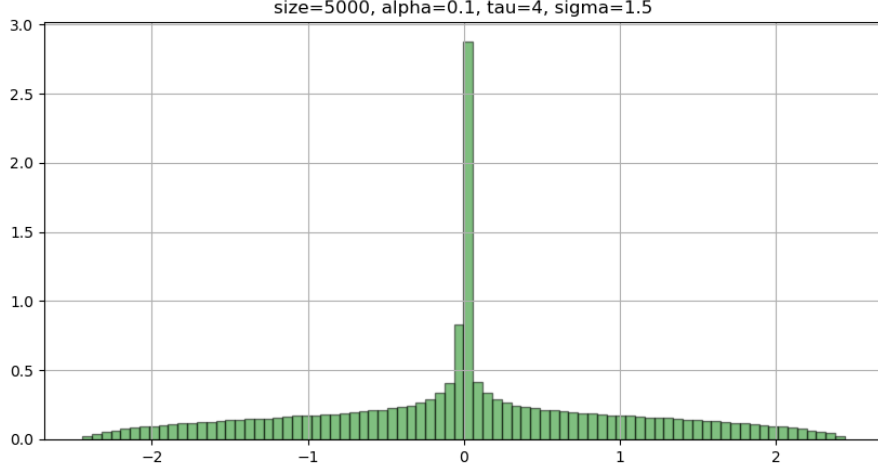
The sufficient universal set alone therefore contributes about 477 forced eigenvalues, or 9.54% of the ESD. Direct evaluation of the source scaling

$$c_N = 2 \sum_{r=1}^{2499} r^{-0.1} + 2500^{-0.1}$$

places them at

$$-c_N^{-1/2} = -0.01984432644 \dots,$$

inside the central histogram bin.



This match is parameter-free beyond the paper’s assumptions: the saturated class depends on α, τ, d but not on the detailed value of $\sigma > 0$.

5 Scope, verification, and novelty

The theorem isolates an exact invariant eigenspace, sharply counts it, and thereby proves that high-connectivity saturation creates the observed kind of central atom. It is a lower bound on the total multiplicity at that point; other graph symmetries may add eigenvectors. Nor does it classify every eigenvalue falling in a finite-width central bin. Such a statement would be a local spectral problem rather than the qualitative conjecture posed by the source.

The calculation was independently checked by a seeded matrix experiment and direct source-parameter evaluation. A bounded exact-title, citation, and keyword search through 11 August 2026 found no later explicit resolution. Because novelty and the interpretation of the informal phrase “due to” remain review-sensitive, the candidate-full label requires human validation.

References

- [1] A. Cipriani, R. S. Hazra, N. Malhotra, and M. Salvi, *The spectrum of dense kernel-based random graphs*, arXiv:2502.09415v2 (2025).